

NETVISOR Mini Discovery V1

Project Information

Project Name: NETVISOR

Module: Discovery Engine

Version: V1

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Date: 2025

1. Project Overview

NETVISOR is a network monitoring and infrastructure management platform designed to discover, monitor, and analyze network assets.

Version 1 focuses on building the first component of the platform:

Mini Discovery Engine

The objective is to discover active devices within a single local network (Single Network Environment) and create a clean foundation for future development.

2. Objectives

The main objectives of Version 1 are:

- Discover active hosts on a local subnet
 - Detect reachable devices using ICMP (Ping)
 - Build a reusable architecture
 - Create a Device model
 - Separate scanning logic from business logic
 - Prepare the system for future upgrades
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3. Architecture

The project follows a modular architecture.

Directory Structure

netvisor-mini-discovery-v1/

core/

- Network.php
- Scanner.php

modules/

- Ping.php

services/

- DeviceBuilder.php

models/

- Device.php

run.php

4. Components

4.1 Device Model

Represents a discovered asset.

Attributes:

- IP Address
- MAC Address (future)
- Status

Purpose:

Provide a unified representation of network assets.

4.2 Network Module

Responsibilities:

- Detect local IP
- Determine network range
- Generate IP addresses for scanning

Functions:

- `getLocalNetwork()`
 - `generateIPs()`
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4.3 Ping Module

Responsibilities:

- Send ICMP Echo Requests
- Detect active hosts

Functions:

- `pingHost()`
-

4.4 Scanner Core

Responsibilities:

- Coordinate discovery process
- Execute ping sweep
- Return active IP addresses

Functions:

- `scanNetwork()`
-

4.5 Device Builder

Responsibilities:

- Convert raw scan results into Device objects

Functions:

- `buildDevices()`
-

5. Discovery Workflow

Step 1:

Detect local network.

Example:

192.168.1.0/24

Step 2:
Generate host list.

192.168.1.1 → 192.168.1.254

Step 3:
Execute Ping Sweep.

Step 4:
Collect active hosts.

Step 5:
Create Device objects.

Step 6:
Display results.

6. Features Implemented

Implemented in V1:

- ✓ Single Network Discovery
- ✓ IP Range Generation
- ✓ ICMP Ping Sweep
- ✓ Active Host Detection
- ✓ Modular Architecture

✓ Device Modeling

✓ CLI Execution

7. Current Limitations

Not implemented yet:

✗ MAC Address Detection

✗ ARP Discovery

✗ TCP Port Scanning

✗ Hostname Detection

✗ OS Fingerprinting

✗ Database Storage

✗ Multi-Network Discovery

✗ Topology Mapping

✗ Monitoring Engine

8. Results

The system successfully identifies active devices within the local network and outputs a list of reachable hosts.

Example:

IP: 192.168.1.1

IP: 192.168.1.35

IP: 192.168.1.38

Total Devices: 3

```
moad21@system-1:~/netvisor-mini-discovery-v1$ nano models/Device.php
moad21@system-1:~/netvisor-mini-discovery-v1$ nano core/Network.php
moad21@system-1:~/netvisor-mini-discovery-v1$ nano modules/Ping.php
moad21@system-1:~/netvisor-mini-discovery-v1$ nano core/Scanner.php
moad21@system-1:~/netvisor-mini-discovery-v1$ nano services/DeviceBuilder.php
moad21@system-1:~/netvisor-mini-discovery-v1$ nano run.php
moad21@system-1:~/netvisor-mini-discovery-v1$ php run.php
NETVISOR MINI DISCOVERY V1
```

===== DEVICES FOUND =====

IP: 192.168.1.2		MAC: unknown		STATUS: online
IP: 192.168.1.15		MAC: unknown		STATUS: online
IP: 192.168.1.27		MAC: unknown		STATUS: online
IP: 192.168.1.42		MAC: unknown		STATUS: online
IP: 192.168.1.56		MAC: unknown		STATUS: online
IP: 192.168.1.68		MAC: unknown		STATUS: online
IP: 192.168.1.76		MAC: unknown		STATUS: online
IP: 192.168.1.77		MAC: unknown		STATUS: online
IP: 192.168.1.80		MAC: unknown		STATUS: online
IP: 192.168.1.91		MAC: unknown		STATUS: online
IP: 192.168.1.105		MAC: unknown		STATUS: online
IP: 192.168.1.107		MAC: unknown		STATUS: online
IP: 192.168.1.110		MAC: unknown		STATUS: online
IP: 192.168.1.123		MAC: unknown		STATUS: online
IP: 192.168.1.130		MAC: unknown		STATUS: online
IP: 192.168.1.157		MAC: unknown		STATUS: online
IP: 192.168.1.254		MAC: unknown		STATUS: online

TOTAL: 17

9. Future Roadmap

Version 2

- ARP Discovery
- MAC Address Collection
- Device Enrichment

Version 3

- TCP Service Detection
- Port Scanning

Version 4

- Multi-Network Discovery

Version 5

- Database Integration

Version 6

- Monitoring Engine

Version 7

- Topology Engine

Version 8

- NETVISOR Dashboard
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10. Conclusion

NETVISOR Mini Discovery V1 establishes the foundation of the NETVISOR platform.

The version successfully performs basic network discovery and provides a clean architecture that supports future expansion toward a complete network monitoring and infrastructure management solution.